

$x = 10$
 $y = 2$

$\text{print}(x+y) \rightarrow 10+2 = \underline{12}$

0^n

$3+3$	$\rightarrow 6$
$3+6$	$\rightarrow 9$
$9+6$	$\rightarrow 15$
$15+15$	$\rightarrow 30$

Input:-

$\text{name} = \text{shaurya} / \text{Armaan} / \text{Your-Mom}$

$\text{input}()$

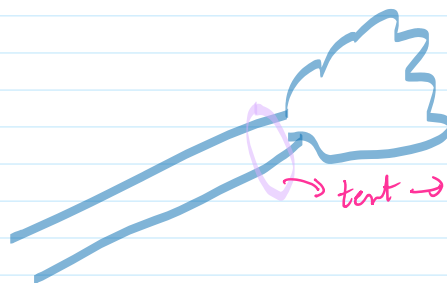
$\rightarrow \text{print} \rightarrow \text{Output on the screen}$

$\rightarrow \text{input} \rightarrow \text{to take input from the user.}$

Band Name Generator:-

- 1) Name $\rightarrow \text{input}$
- 2) Your-favourite pet $\rightarrow \text{input}$
- 3) Your-favourite city $\rightarrow \text{input}$

$\text{input}()$



$\rightarrow \text{tent} \rightarrow \text{Name} + \text{favourite pet} + \text{favourite city}$

"Shaurya" → string

"12a Shaurya" → string
↓ → int

Data Types

Imagine a bank account.

A bank account contains different kinds of information:

- Account Holder Name → "Shanmuka" → Shaurya-Hon
- Account Balance → 50000.75
- Account Active → True
- Transaction IDs → [101, 102, 103] → Collection

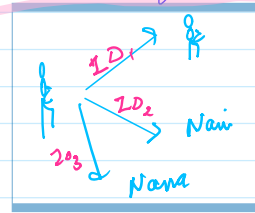
All these values are different.

Python needs to know:

- Which value is text? → Name name
- Which value is a number? → Balance
- Which value is a yes/no answer? → Account
- Which value is a collection? → Trans.

This is where **Data Types** become important.

What kind of data it stores → data type



Data Types

Single Value

Collection Data Types

These store one value at a time.

Examples:

- int → age → 12
- float → wt → 70.59
- str → name → Tejas
- bool → is active → false

True/Yes No/False

These store multiple values together.

Examples:

- list
- tuple
- set
- dict

leave

that can change

Data Type	Example	Mutable/Immutable	Ordered/Unordered	Real-Life Example
int	100	Immutable	N/A	Roll Number
float	99.5	Immutable	N/A	Product Price
str	"Python"	Immutable	Ordered	Student Name
bool	True	Immutable	N/A	Login Status
list	[1,2,3]	Mutable	Ordered	Shopping Cart
tuple	(1,2,3)	Immutable	Ordered	Coordinates
set	{1,2,3}	Mutable	Unordered	Unique IDs
dict	{'name': 'Rahul'}	Mutable	Insertion-Ordered	Student Record